Noise Injection Strategies in Deep Q-Networks: Addressing TD Loss Collapse and Exploration in Sparse Environments

Abstract

Standard ϵ-greedy Deep Q-Networks (DQNs) often fail in sparse-reward environments due to dithering and premature zero Temporal Difference (TD) loss collapse, which destroys network plasticity. This report formalizes the mathematical impact of injecting noise into the Q-network's feature space. We derive the expected noisy loss bounds for both additive and Hadamard (Gaussian Dropout) noise perturbations, explicitly decomposing the implicit regularization and bias terms. We then address the underfitting side effects of the Hadamard approach and provide structural mitigations. Finally, we provide a comparative analysis between these manual noise strategies and DeepMind's NoisyNets.

1 Introduction

In off-policy Reinforcement Learning, the standard TD loss optimized during training is $L_{clean} = (Y - Q_a)^2$, where $Y = r + \gamma \max_{a'} Q'_b(a')$ and $Q_a = \phi(s)^T w_a$. In sparse environments, the target Y frequently evaluates to zero, driving $Q_a \to 0$ and $L_{clean} \to 0$. This gradient starvation leads to a loss of plasticity.

Injecting random noise into the feature representation $\phi(s)$ forces the network to optimize a modified objective, maintaining gradient flow and driving deep exploration. We analyze two formulations: additive noise and Hadamard noise.

2 Derivation of Noisy Loss: Additive Case

Let the injected feature noise be $m \sim \mathcal{N}(0, \sigma^2 I)$. The online noisy Q-value is $Q_{a,m} = (\phi(s)+m)^T w_a$. Similarly, the noisy target evaluates to $Y_m = r + \gamma \max_{a'} (\phi(s') + m')^T w_b(a')$.

The expected noisy TD loss is:

$$\mathbb{E}[L_{noisy}] = \mathbb{E}_{m,m'} \left[(Y_m - Q_{a,m})^2 \right] \tag{1}$$

By adding and subtracting the clean target Y, and substituting the clean TD error $\delta = Y - Q_a$, we expand the polynomial:

$$\mathbb{E}[L_{noisy}] = \mathbb{E} \left[((Y_m - Y) + (Y - Q_{a,m}))^2 \right] \tag{2}$$

$$= \mathbb{E} \left[((Y_m - Y) + \delta - m^T w_a)^2 \right] \tag{3}$$

Expanding the square and applying the expectations (noting $\mathbb{E}[m] = 0$ and m, m' are independent):

$$\mathbb{E}[L_{noisy}] = \underbrace{\delta^2}_{L_{clean}} + \underbrace{2\delta \mathbb{E}[Y_m - Y]}_{\text{Bias}} + \underbrace{\mathbb{E}[(Y_m - Y)^2]}_{\text{Target Variance}} + \underbrace{\mathbb{E}[(m^T w_a)^2]}_{\text{Online Variance}} \tag{4}$$

Because $\text{Cov}(m) = \sigma^2 I$, the online variance simplifies to $\sigma^2 \|w_a\|_2^2$. Bounding the target variance using the property of the maximum operator yields the upper bound:

$$L_{noisy} \leq L_{clean} + \sigma^2 \left(\|w_a\|_2^2 + \gamma^2 \sum_{a'} \|w_b(a')\|_2^2 \right) + 2\gamma\sigma|\delta| \sqrt{\sum_{a'} \|w_b(a')\|_2^2} \tag{5}$$

3 Derivation of Noisy Loss: Hadamard Case

To apply state-dependent variance, we utilize a Hadamard product $m = \mathbf{1} + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$.

The noisy feature vector becomes $\phi(s) \odot (\mathbf{1} + \epsilon) = \phi(s) + \phi(s) \odot \epsilon$. The noisy Q-value calculation is:

$$Q_{a,m} = (\phi(s) + \phi(s) \odot \epsilon)^T w_a = \phi(s)^T w_a + (\phi(s) \odot \epsilon)^T w_a \tag{6}$$

By rearranging the dot product, the perturbation becomes $\epsilon^T (\phi(s) \odot w_a)$.

Following the exact expansion steps from the additive case, but substituting w_a with $(\phi(s) \odot w_a)$, we arrive at the expected decomposed loss:

$$\mathbb{E}[L_{noisy}] = L_{clean} + 2\delta \mathbb{E}[Y_m - Y] + \mathbb{E}[(Y_m - Y)^2] + \sigma^2 \|\phi(s) \odot w_a\|_2^2 \tag{7}$$

And the corresponding upper bound inequality:

$$L_{noisy} \leq L_{clean} + \sigma^2 \left(\sum_i (\phi_i(s) w_{a,i})^2 + \gamma^2 \sum_{a',i} (\phi_i(s') w_{b,i}(a'))^2 \right) + 2\gamma\sigma|\delta| \sqrt{\sum_{a'} \|\phi(s') \odot w_b(a')\|_2^2} \tag{8}$$

3.1 Significance of the Hadamard Loss Terms

The decomposed expected loss explicitly reveals how Gaussian Dropout on the feature space fundamentally alters network architecture and learning dynamics:

- **Online Variance Penalty $\left(\sigma^2 \sum \phi_i(s)^2 w_{a,i}^2\right)$:** This term imposes a strict implicit L_2 regularization that scales quadratically with feature activation.
  - *Feature Correlation Regularization:* By penalizing the reliance on any single massive feature activation $\phi_i(s)$, the network is forced to distribute the Q-value estimation across a broader, uncorrelated set of features, improving representation robustness.
  - *Preventing TD Loss Collapse:* Even if $L_{clean} \to 0$, this variance term ensures $\mathbb{E}[L_{noisy}] > 0$. Gradients persistently flow through active pathways, completely circumventing zero TD loss collapse and maintaining network plasticity in sparse environments.
  - *Underfitting Risk:* If σ^2 is too high, this term acts as a crushing weight decay on active features, forcing the optimizer to underestimate Q-values.
- **Target Variance & Bias $\left(\mathbb{E}[(Y_m - Y)^2]$ and $2\delta\mathbb{E}[Y_m - Y]\right)$:** By Jensen's Inequality, target noise combined with the max operator creates a strictly positive overestimation bias. This structurally implements "Optimism in the Face of Uncertainty," intrinsically motivating deep exploration.

3.2 Mitigating Underfitting in the Hadamard Formulation

Because the implicit regularization scales quadratically with both the feature activation and the weight ($\sigma^2\phi_i(s)^2 w_{a,i}^2$), learning high Q-values forces a massive variance penalty, resulting in severe underfitting. This mathematical link can be broken or constrained through the following algorithmic improvements:

- **Off-Policy Decoupling (Action-Only Noise):** Because DQNs learn off-policy, exploration noise does not need to enter the optimization objective. By applying Hadamard noise exclusively during environment interaction (rollout) and utilizing clean features during replay buffer optimization, the variance penalty is entirely removed from the gradient backward pass, allowing weights to grow unobstructed.
- **Dual-Noise Strategy:** To retain architectural plasticity without causing underestimation, two separate variance scales are employed: a large $\sigma_{explore}^2$ during action selection for deep exploration, and a tightly constrained σ_{train}^2 during the optimization pass to provide non-destructive regularization.
- **Layer Normalization:** Applying Layer Normalization immediately prior to the noise injection bounds the variance of the features to unity. This places a hard mathematical ceiling on $\phi_i(s)^2$, transforming the dynamically exploding variance penalty into a stable, predictable weight decay term driven purely by $w_{a,i}^2$.
- **Reward Scaling:** Explicitly normalizing environment rewards down limits the maximum theoretical Q-value to $\frac{R_{max}}{1-\gamma}$. This allows the network to approximate targets using small weight magnitudes, keeping the multiplicative variance penalty structurally negligible.

4 Comparative Discussion: Noise Strategies

Three distinct strategies exist for feature and parameter perturbation, each with structural trade-offs.

4.1 Additive Feature Noise

The additive approach ($\phi(s) + \epsilon$) unconditionally adds variance uniformly across all dimensions.

- **Pros:** Computationally trivial; guarantees continuous plasticity.
- **Cons:** State-agnostic. It corrupts empty regions of the feature manifold ($\phi_i = 0$), leading to chaotic Q-value updates and high risk of TD loss explosion as weight norms grow unconstrained.

4.2 Hadamard Feature Noise (Gaussian Dropout)

The Hadamard approach ($\phi(s) \odot (1 + \epsilon)$) scales noise by feature activation.

- **Pros:** Phenomenal for diagnosing and correcting architectural flaws like representation collapse and highly correlated state transitions. Zero parameter overhead.
- **Cons:** Exploration variance and weight decay are inextricably mathematically coupled, requiring the aforementioned mitigation techniques or explicit decay schedules to allow the policy to converge.

4.3 NoisyNets (DeepMind)

NoisyNets replace standard linear parameters with $w = \mu_w + \sigma_w \odot \epsilon$, injecting learnable noise directly into the weight space.

- **Pros:** *Auto-annealing.* The optimizer can minimize the noisy TD loss by shrinking σ_w specifically in states it has mastered. It successfully decouples the exploration scale (σ_w) from the feature importance (μ_w), enabling highly stable, long-horizon deep exploration without manual hyperparameter tuning.
- **Cons:** Doubles the parameter count of affected layers; structurally complex to implement efficiently; can still suffer from premature convergence (shrinking $\sigma_w \to 0$ too early) during long streaks of zero rewards.

5 Conclusion

In sparse environments where target Q-values frequently evaluate to zero, standard DQNs lose plasticity and succumb to dithering. While NoisyNets provide the most hands-off solution for exploration, the Hadamard formulation provides superior, mathematically guaranteed regularization of feature correlations and safeguards against zero TD loss collapse without requiring additional network parameters, provided structural modifications like Off-Policy Decoupling or Layer Normalization are employed to prevent underfitting.